AI-Driven Data Analytics Curriculum – Version 8

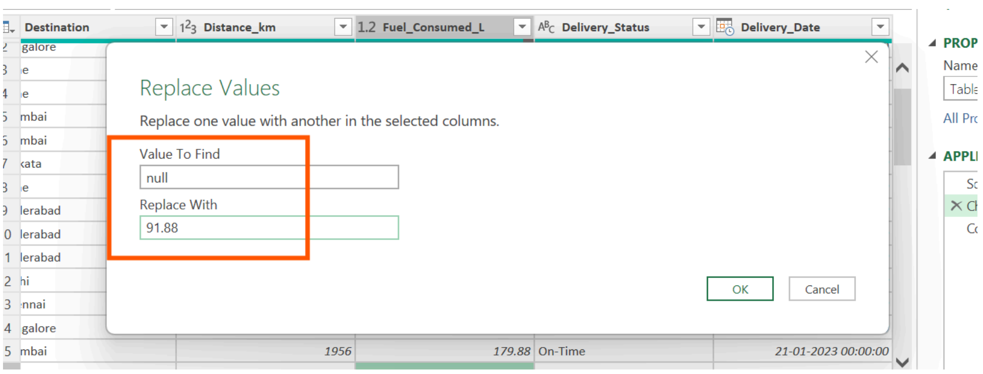
Power BI Module End Assignment

Title: AI-Powered Fleet Performance & Delivery Efficiency Dashboard

1. Data Cleaning & Modeling:

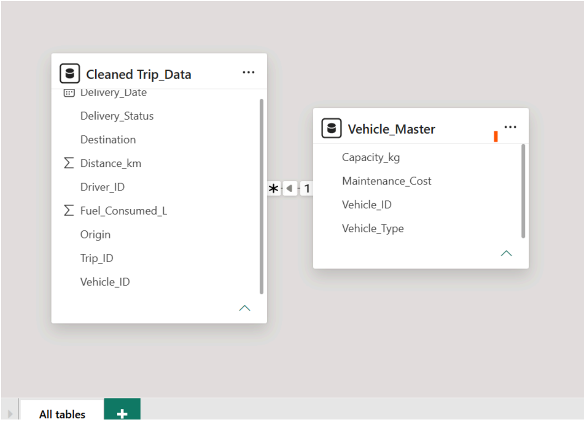
○ Fix missing fuel consumption values (use mean imputation).

First, I converted the dataset into Power Query. Then, I used Transform → Statistics → Average to find the mean of Fuel Consumption. The average value was 91.88. Finally, I used Replace Values to replace the null values with 91.88.



○ Create a relationship using Vehicle_ID between Trip_Data and the Vehicle Master table.

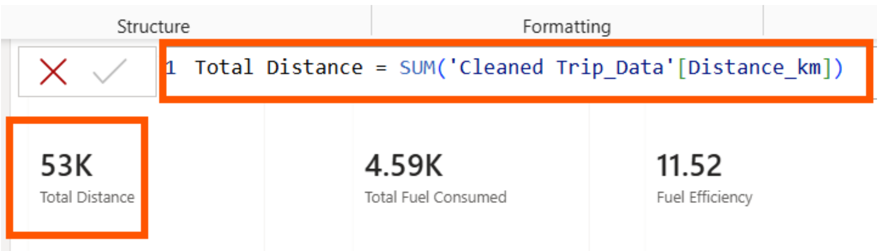
Vehicle Master table has 8 rows of unique vehicle id's, and Cleaned Trip Data table has 50 trips. One vehicle can have many trips. So the relationship is One-to-Many (1:*)

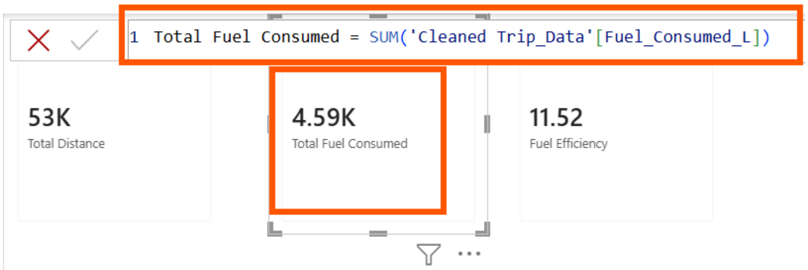


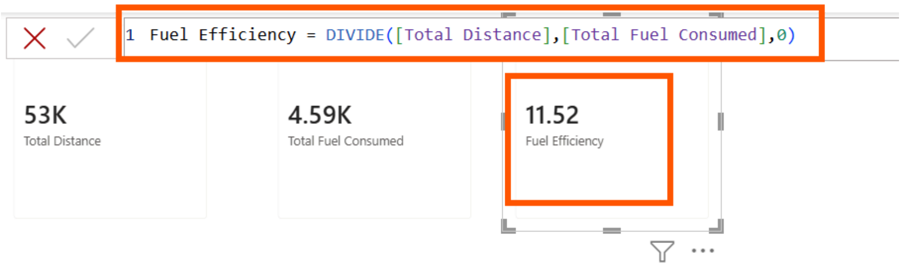
2. DAX Measures:

○ Fuel Efficiency = Distance / Fuel Consumed

First I have calculated Total Distance and Total Fuel Consumed. Then I have calculated Fuel Efficiency using Total Distance / Total Fuel Consumed

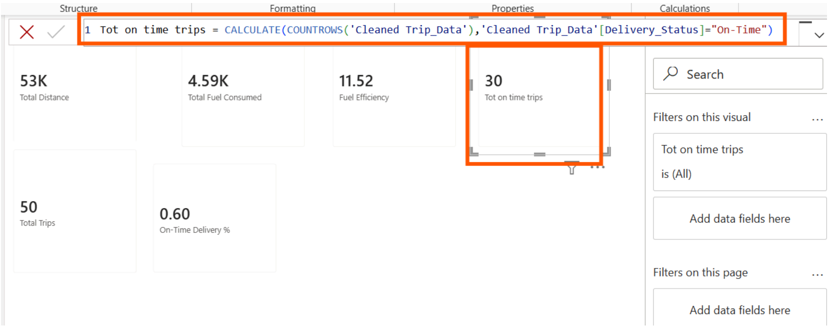


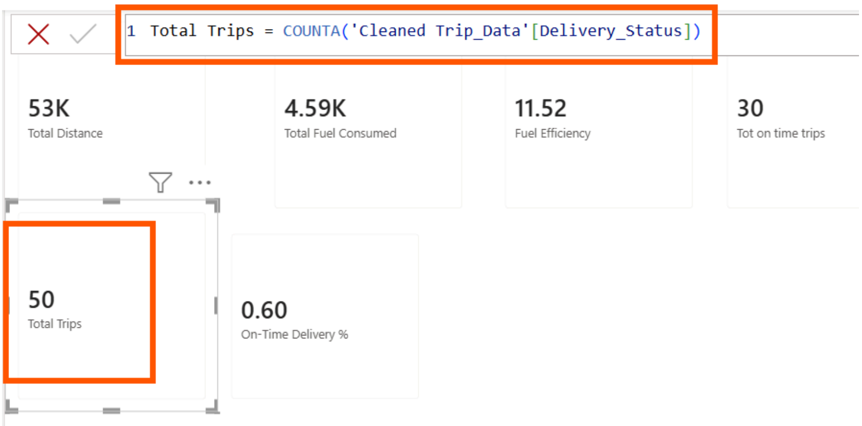


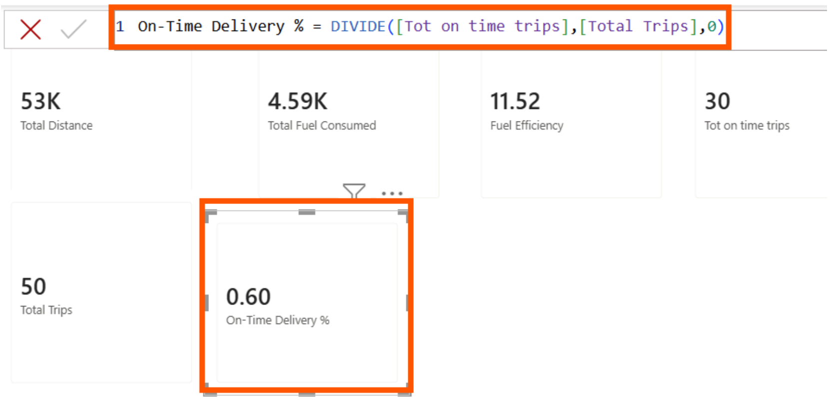


○ On-Time Delivery % = On-Time Trips / Total Trips:

First I have counted the On-Time Trips and Total Trips. Then I have divided On-Time Trips by Total Trips to calculate the On-Time Delivery %.

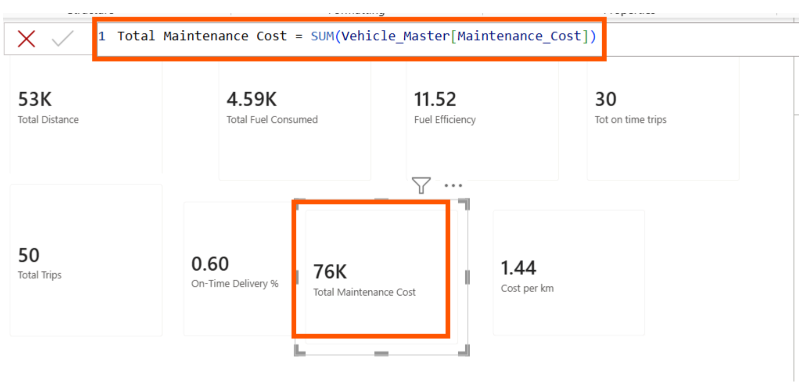


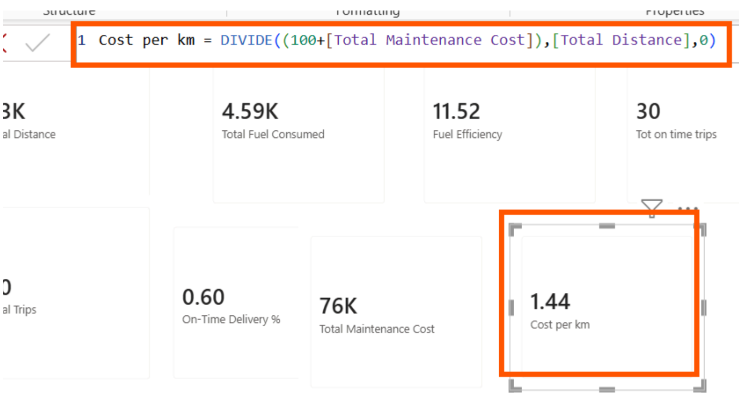




○ **Cost per km = (fuel cost + Maintenance Cost) / Distance**
Note fuel cost =100

First I have calculated Total Maintenance Cost. Then I used the formula (Fuel Cost + Maintenance Cost) / Distance. Finally I have calculated the Cost per km.





3. Visualization:

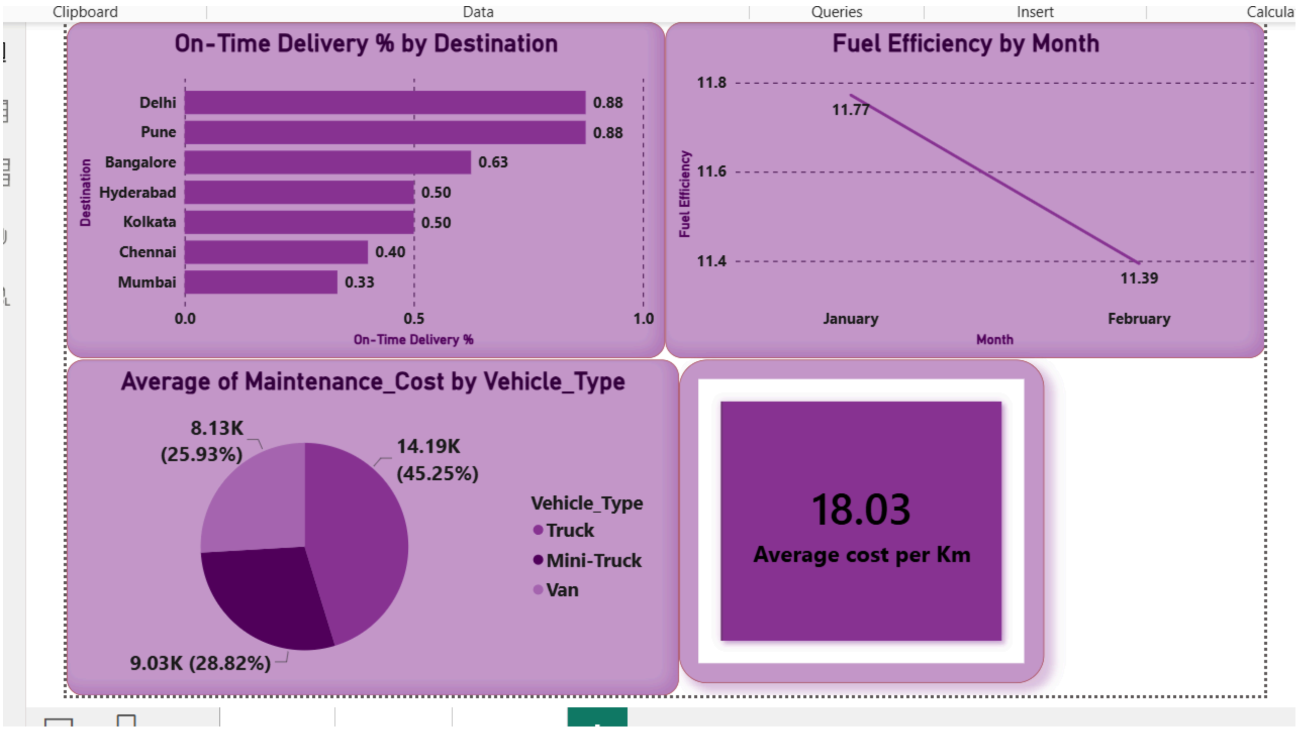
- ○ Bar chart: On-Time Delivery % by Destination.
- ○ Line chart: Fuel efficiency trend by delivery date.
- ○ Cards visualization:
  - ■ Avg. Delivery Time
  - ■ Average cost per km
- ○ Pie chart: vehicle type vs Average maintenance cost

I created a bar chart to show On-Time Delivery % by Destination.

I created a line chart to show the Fuel Efficiency trend by Delivery Date.

Average Delivery Time cannot be calculated because Order Date is not available.

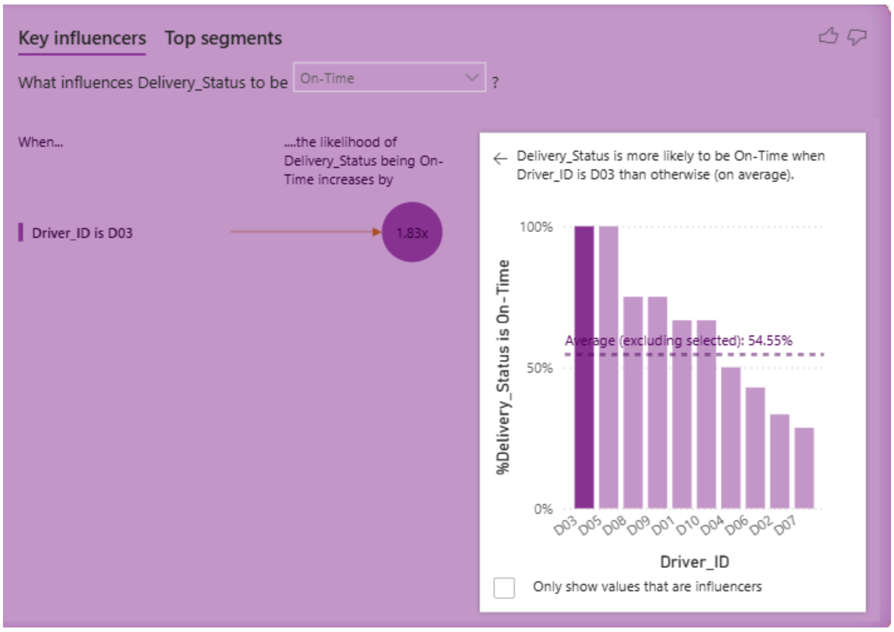
I created a Card to show the Average Cost per km, and the value is 18.03.



4. AI-Powered Visuals

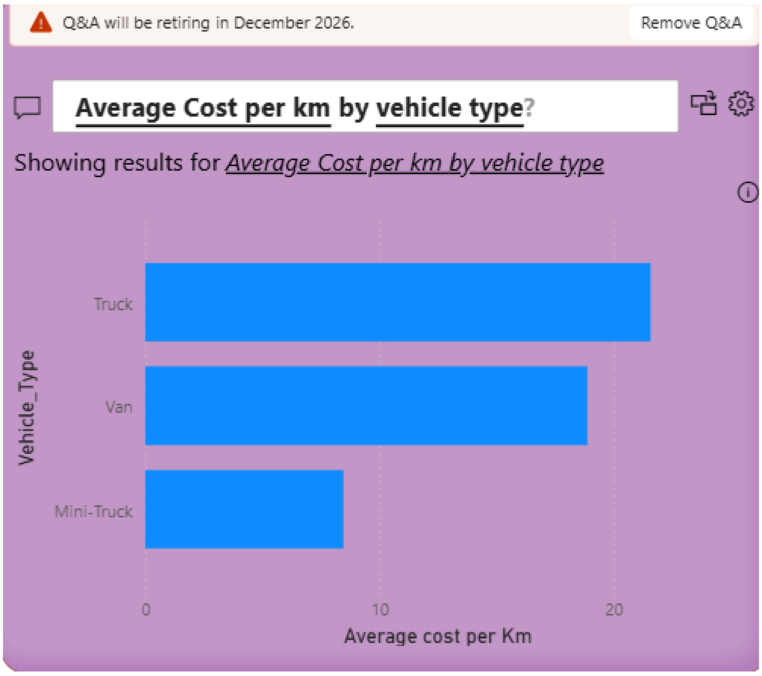
❖ Key Influencers Visual: ➢ Add Delivery Status in the Analyze and explain by -distance in km,vehicle_type, Driver_ID

The Key Influencers visual shows that D03 has better On-Time delivery performance.



❖ Q&A Visual: ➢ Add Q&A visual → Prompt: “Average Cost per km by vehicle type?”

I have created the Q&A visual.It shows the average cost per km by vehicle type



❖ Decomposition Tree (AI Visual): ➢ Analyze Cost per km → explained by Vehicle Type, Maintenance_cost, and Distance_km.

I have Created the Decomposition Tree to analyze Cost per km by Vehicle Type, Maintenance Cost, and Distance km